

TOP COMMAND – MANUAL NOTES

Command Used:

Top

This command is used to check **system performance (CPU, Memory, Processes)** in real time

1. FIRST LINE

top - 10:50:51 up 12:33, 2 users, load average: 0.03, 0.03, 0.00

Notes:

- **10:50:51** → Current system time
- **up 12:33** → System 12 hours 33 minutes se ON hai
- **2 users** → 2 users login hai system me

Load Average:

- **0.03** (1 min)
- **0.03** (5 min)
- **0.00** (15 min)

Load ka matlab = Load Average batata hai ki CPU kitna busy hai aur kitne processes CPU ka wait kar rahe hain.

Agar load < CPU cores → system free hai

Agar load zyada ho → system slow ho sakta hai

Yaha load bahut low hai → system idle hai

2. TASKS (PROCESSES)

Tasks: 44 total, 1 running, 43 sleeping, 0 stopped, 0 zombie

🔑 Notes:

- **44 total** → Total processes system me chal rahe hai

- **1 running** → sirf 1 process active hai
- **43 sleeping** → baaki process wait kar rahe hai
- **0 stopped** → koi process pause nahi hai
- **0 zombie** → A zombie process is a terminated process whose exit status is not yet collected by its parent process.

Zombie process dangerous hota hai (yaha safe hai)

%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st

Notes:

- **Us (user)** → user applications CPU use kar rahe hai
- **Sy (system)** → OS (kernel) CPU use kar raha hai
- **ni (nice)** → low priority process
- **id (idle)** → CPU free hai
- **wa (I/O wait)** → CPU deta ka intajaar kiya ja raha hai (disk/network se).
- **Hi (Hardware Interrupt)** → CPU apana kitana time hardware device ke signals handle karane mein laga raha hai
- **Si (software interrupt)** → CPU apana kitana time software device ke signals handle karane mein laga raha hai
- **St (Steal Time)** → yah hypervisor dvaara any virtual machine ko chalaane ke lie upayog kie gae cpu samay ke % ko darshaata hai.

Yaha:

- CPU 99.7% idle hai → system bilkul free hai
-

4. MEMORY (RAM)

MiB Mem : 3751.2 total, 3003.6 free, 508.2 used, 329.6 buff/cache

Notes:

- **Total RAM** → 3751 MB (~4GB)
- **Free** → 3003 MB (bahut free hai)
- **Used** → 508 MB
- **buff/cache** → 329 MB (temporary storage for speed)

System me RAM bhi kaafi free hai

5. SWAP MEMORY

MiB Swap: 1024.0 total, 1024.0 free, 0.0 used

Notes:

- **Swap** = backup memory (hard disk ka use)
- **Total** = 1GB
- **Used** = 0 → swap use nahi ho raha

Good sign (RAM enough hai)

6. PROCESS TABLE (DETAILS)

Columns:

- **PID** → Process ID
 - **USER** → Kis user ne process start kiya
 - **PR** → Priority
 - **NI** → Nice value (priority control)
 - **VIRT** → Total virtual memory
 - **RES** → Actual RAM used
 - **SHR** → Shared memory
 - **S** → State (Running/Sleeping)
 - **%CPU** → CPU usage
 - **%MEM** → Memory usage
 - **TIME+** → Total CPU time
 - **COMMAND** → Process name
-

Example Processes:

1. containerd

- Docker/container service
- CPU: 1.0%
- Memory: 1.4%

👉 Containers handle karta hai

2. snap

- Snap packages manage karta hai
 - CPU: 0.7%
-

3. systemd-udevd

- Hardware devices manage karta hai
-

4. systemd (PID 1)

- Main process of Linux OS
 - Sab process isi se start hote hai
-

5. init

- System initialization process
-

FINAL CONCLUSION

- ✓ CPU almost free (99.7% idle)
- ✓ Load very low (0.03)
- ✓ RAM mostly free
- ✓ Swap not used
- ✓ No zombie process

👉 **System health = Very Good** 100

SHORT SUMMARY

- System idle hai
- Koi heavy load nahi hai
- Performance smooth hai
- Server safe & stable hai



Note:

Ye output ek **healthy Linux system** ka example hai.